

Artificial Intelligence in Pediatric Radiology

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Objectives

- Describe the big players in radiology AI, both general and pediatric, since 2023
- Summarize what radiology AI does well and what remains unresolved

Methods

Source	How the data were collected	How pediatric was subset
Journals, preprints	PubMed yearly counts, plus arXiv and medRxiv (Europe PMC) with the same query. Most-cited papers: Semantic Scholar, ranked by citations. Systematic review: PubMed + Embase, 2005–present, every record screened and read (3,496 primary studies).	Counts: radiology AND AI, then AND pediatric (query below). Review: pediatric query, then screened for a pediatric population. Most-cited lists: the title must also contain a pediatric term (as for conferences).
Conferences	Semantic Scholar venue search, 2023–present: NeurIPS, ICLR, ICML, CVPR, MICCAI, MIDL (radiology terms, then a title filter); RSNA and SPR through their journals (AI terms), since meeting abstracts are not indexed.	Title contains a pediatric term (pediatric, paediatric, child, infant, neonat, adolescent, fetal, foetal, newborn, bone age, pediatrics, paediatrics, kawasaki, scoliosis, preterm, premature, congenital, young adult); marked with a star.
Datasets	Hand-curated list of public imaging datasets; sizes checked against the primary paper or hosting page.	Pediatric by selection: only datasets of children or fetuses are listed.
Newsletters	Full archives of The Imaging Wire, RSNA News, TLDR AI, TLDR Tech, Signify Research, ESR / ECR, Radiology Business, split into individual stories. Radiology AI = an AI term and an imaging term in the same story.	A pediatric term (pediatric, child, infant, neonat, newborn, adolescen, fetal, bone age, preterm, NICU, prenatal, ...) also in the same story.
Commercial	FDA list of AI-enabled devices, Radiology panel (all ages).	Curated products with a stated pediatric indication or pediatric-by-design use, cross-checked against the FDA list.

PubMed query = radiology AND AI AND pediatric (systematic review corpus, 2005–present; translated term for term for Embase; [tiab] = title/abstract)

radiology: (radiology[tiab] OR radiological[tiab] OR radiograph[tiab] OR "medical imaging"[tiab] OR "diagnostic imaging"[tiab] OR tomography[tiab] OR "magnetic resonance"[tiab] OR MRI[tiab] OR "MR imaging"[tiab] OR "MR images"[tiab] OR MRIs[tiab] OR "MR image"[tiab] OR "MR scan"[tiab] OR "MR scans"[tiab] OR "computed tomography"[tiab] OR CT[tiab] OR ultrasound[tiab] OR ultrasonograph[tiab] OR sonograph[tiab] OR echocardiograph[tiab] OR mammograph[tiab] OR "chest x-ray"[tiab] OR "x-ray"[tiab] OR fluoroscop[tiab] OR scintigraph[tiab] OR "positron emission"[tiab] OR PET[tiab] OR SPECT[tiab] OR angiograph[tiab] OR neuroimaging[tiab] OR tractograph[tiab] OR elastograph[tiab] OR "fMRI"[tiab] OR "functional MRI"[tiab] OR "functional magnetic resonance"[tiab] OR connectome[tiab] OR "diffusion tensor"[tiab] OR DT[tiab] OR "bone age"[tiab] OR "skeletal age"[tiab] OR "skeletal maturity"[tiab] OR "barium enema"[tiab])

AI: ("artificial intelligence"[tiab] OR "machine learning"[tiab] OR "deep learning"[tiab] OR "convolutional neural network"[tiab] OR "convolutional neural networks"[tiab] OR "neural network"[tiab] OR "neural networks"[tiab] OR "computer-aided diagnosis"[tiab] OR "computer aided diagnosis"[tiab] OR "computer-aided detection"[tiab] OR radiomic[tiab] OR "computer vision"[tiab] OR "large language model"[tiab] OR "large language models"[tiab] OR "foundation model"[tiab] OR "foundation models"[tiab] OR "transfer learning"[tiab] OR "vision transformer"[tiab] OR "self-supervised"[tiab] OR "random forest"[tiab] OR "support vector machine"[tiab] OR "gradient boosting"[tiab] OR "natural language processing"[tiab])

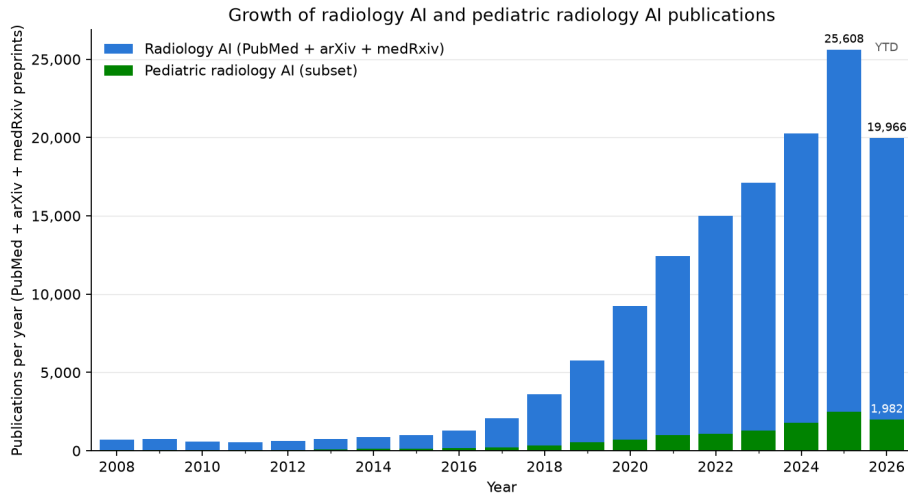
pediatric: (pediatric[tiab] OR paediatric[tiab] OR child[tiab] OR infant[tiab] OR neonat[tiab] OR adolescen[tiab] OR fetal[tiab] OR foetal[tiab] OR fetus[tiab] OR foetus[tiab] OR newborn[tiab] OR preterm[tiab] OR "premature infant"[tiab] OR "premature infants"[tiab] OR "premature birth"[tiab] OR "premature neonate"[tiab] OR "premature neonates"[tiab] OR prenatal[tiab] OR antenatal[tiab] OR perinatal[tiab] OR youth[tiab] OR juvenile[tiab] OR "children's hospital"[tiab] OR schoolchild[tiab] OR toddler[tiab])

Publication-count charts (radiology AI = radiology AND AI; pediatric = AND pediatric):

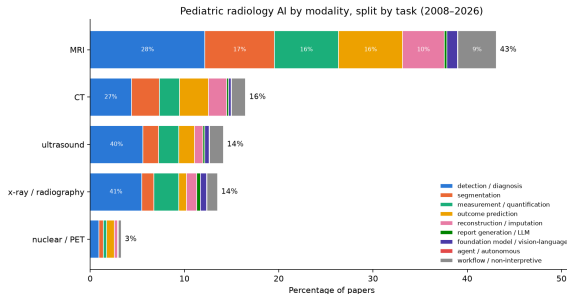
radiology: ((radiology OR radiological OR radiograph OR "medical imaging" OR "diagnostic imaging"[tiab] OR "tomography, x-ray computed"[MeSH Terms] OR "tomography, emission-computed"[MeSH Terms] OR tomograph[tiab] OR "magnetic resonance" OR MRI OR "computed tomography" OR CT[tiab] OR ultrasound[tiab] OR ultrasonograph[tiab] OR sonograph[tiab] OR echocardiograph[tiab] OR "ultrasonography"[MeSH Terms] OR mammograph OR "chest x-ray") NOT ("optical coherence"[tiab] OR fundus[tiab] OR retina[tiab] OR dental[tiab] OR histopatholog[tiab] OR "whole slide"[tiab] OR dermoscop[tiab] OR endoscop[tiab] OR colonoscop[tiab] OR microscop[tiab]))

AI: ("artificial intelligence" OR "machine learning" OR "deep learning" OR "convolutional neural network" OR "neural network" OR "computer-aided diagnosis" OR radiomics OR "computer vision")

pediatric: (pediatric OR paediatric OR child OR infant OR neonat OR adolescen OR "children's hospital")



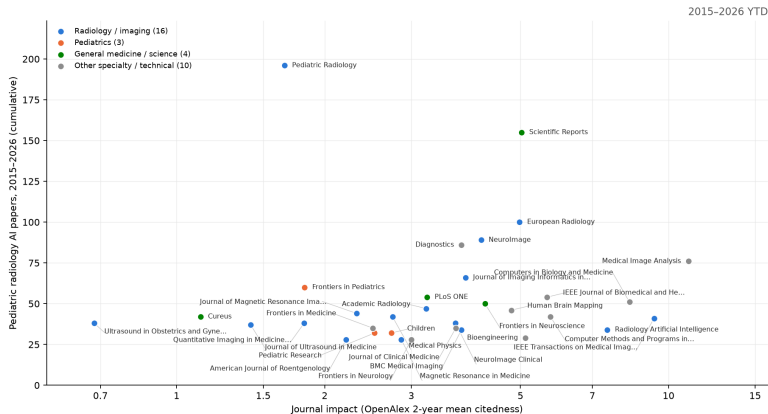
Pediatric radiology AI publications, stratified by modality and task



Same construction, restricted to the pediatric subset (10,737 papers incl. 596 preprints). Top pediatric modalities: MRI 41%, CT 16%, ultrasound 13%, x-ray / radiography 12%. Mammography is omitted: its 79 pediatric-query hits (of 10,737) are adult breast-imaging papers whose abstracts mention children.

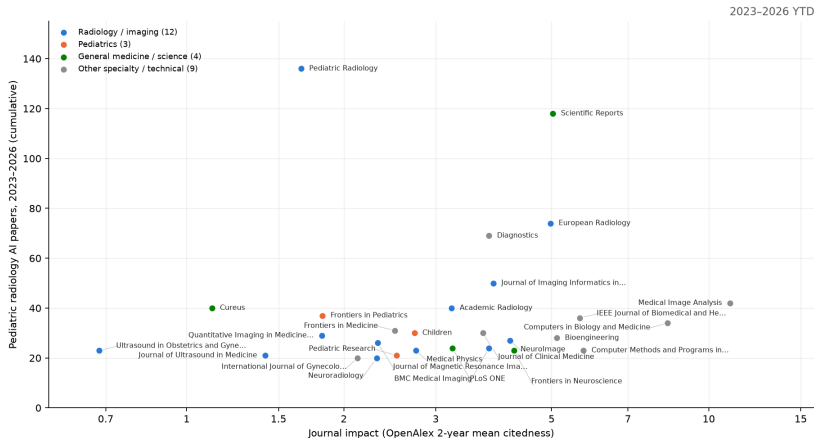
- detection / diagnosis:** is a finding present now, and which one (label or box): fracture, pneumonia, tumor; includes triage and screening
- segmentation:** outline a structure or lesion (a mask)
- measurement / quantification:** a number from the image: bone age, Cobb angle, organ volume, fetal biometry
- outcome prediction:** a future risk, response, or survival estimate from the image (prognosis, radiomics signatures)
- reconstruction / imputation:** a better or missing image: lower dose, faster scans, denoising, synthetic CT/MR
- report generation / LLM:** text: draft, summarize, or extract from radiology reports
- foundation model / vision-language:** a large pretrained model reused across tasks (segment-anything, self-supervised)
- agent / autonomous:** multi-step actions taken without a human in the loop
- workflow / non-interpretive:** protocoling, scheduling, ordering, decision support, education

Which journals publish pediatric radiology AI



x = OpenAlex 2-year mean citedness (a free JIF analogue; runs below the published JIF). y = cumulative pediatric radiology-AI papers 2015-2026 (PubMed, strict title/abstract query). The 33 journals with at least 25 of them are shown, carrying 1,807 of 5,099; the rest is a long tail. Largest: Pediatric Radiology (196); median impact 3.7, 1 at 10 or above – this work lands mostly in mid-impact specialty and technical journals. A further 3 conference proceedings and preprint servers clear the threshold (94 papers) but have no impact number of any kind and are not plotted. The same chart animated year by year: [figures/journal_impact.gif](#).

Which journals publish pediatric radiology AI, 2023–present



x = OpenAlex 2-year mean citedness (a free JIF analogue; runs below the published JIF). y = cumulative pediatric radiology-AI papers 2023-2026 (PubMed, strict title/abstract query). The 28 journals with at least 20 of them are shown, carrying 1,099 of 3,340; the rest is a long tail. Largest: Pediatric Radiology (136); median impact 3.2, 1 at 10 or above – this work lands mostly in mid-impact specialty and technical journals. medRxiv clears the threshold (31 papers) but has no impact number of any kind, so it is not plotted.

Cross-check: an independent 2025 scoping review of pediatric radiology AI

Kamran et al., *Pediatric Radiology* 2025 (doi 10.1007/s00247-025-06462-5): 789 original pediatric-focused articles hand-screened from four databases, 2005 to August 2024. Our PubMed counts are larger because they include any paper that mentions a pediatric term.

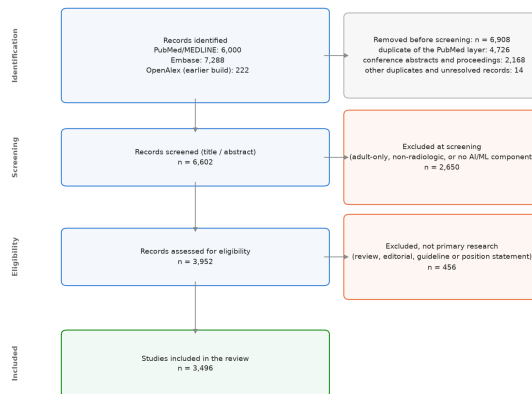
- **Modality:** radiography 38%, MRI 33%, ultrasound 14%, CT 11%, nuclear 2%. Our pediatric query ranks MRI first (MRI 41%, CT 16%, ultrasound 13%, x-ray / radiography 12%) because MRI-heavy neurodevelopment papers mention children.
- **Subspecialty:** musculoskeletal 33%, neuro 29%, chest 17%, body 14%, cardiac 6%.
- **Top applications:** bone age (88 articles), pneumonia (65), brain tumors (54), scoliosis (45), hip dysplasia (28), congenital heart disease (25), autism (18), epilepsy (14), ADHD (9), hydrocephalus (9).
- **Use of AI:** 91% image interpretation / diagnosis; 5.6% image quality; under 2% for acquisition, communication, protocoling, education, and policy combined.
- **Data:** 65% used a single local hospital dataset; 7% did not report the dataset; China 28%, USA 25%, Canada 7% of articles.
- **Growth:** 23 articles in 2018, 158 in 2022, 171 in 2023, consistent with our PubMed curve.
- **Recommendations:** multi-institutional, age-diverse pediatric datasets; CLAIM-style reporting with subgroup results and calibration; funding for the neglected non-interpretive areas (policy, education, implementation); children and families as stakeholders across the AI lifecycle.
- **What it adds to this deck:** the same three leading tasks (bone age, pneumonia, brain tumor) and the same gap (small local datasets, adult tools applied to children) reached by a manual review.

Reading the whole pediatric literature, not a sample of it

Counting publications says how much is being published; it does not say what was built, for which children, or how well it was tested. So every pediatric radiology-AI record from 2005 onward was retrieved from **PubMed/MEDLINE and Embase**, screened against prespecified criteria, and read into a structured row — model, modality, body region, age group, task, dataset size, data source, validation, availability — under one fixed schema.

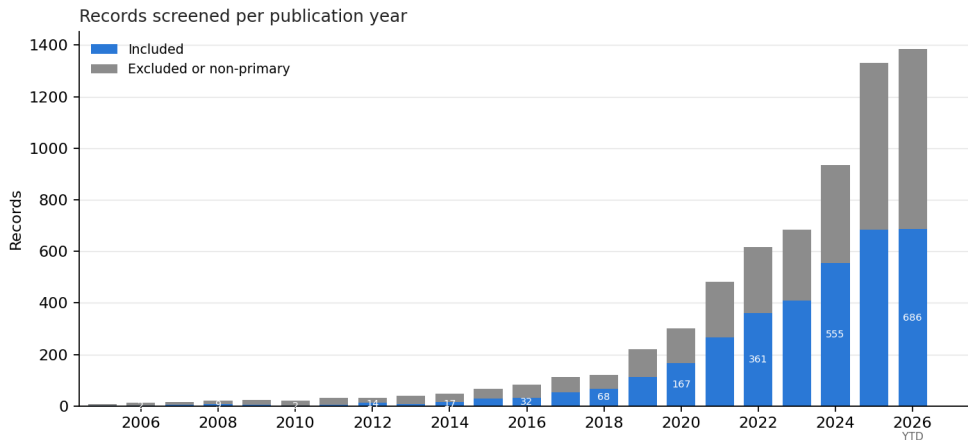
- 6,995 records retrieved, 393 conference proceedings set aside, 6,602 screened, 2,650 off-topic and 456 non-primary removed, 3,496 primary studies included (PubMed 3,122, Embase 365, OpenAlex 9).
- **Embase earns its place:** 365 of the included studies are not in PubMed at all. They have no PMID, so no NIH relative citation ratio; ranking them at all is why the impact column below is OpenAlex's field-weighted score.
- Conference proceedings are retrieved and reported but not screened: an abstract cannot support a judgement about study design, and it carries no DOI linking it to the later full paper.
- Screening and extraction were done by a constrained model under a fixed schema, with a blinded second-model re-screen of 150 records (94.7% agreement, $\kappa = 0.885$). Dual-human validation is outstanding.
- The same corpus is the paper database (`reports/04_paper_database.md`) and the systematic review manuscript, so the three never disagree.

Study selection



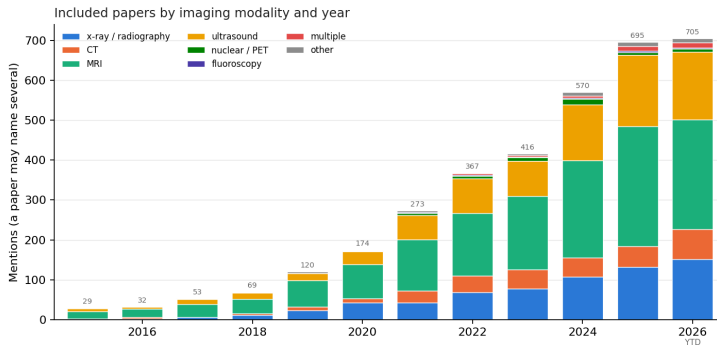
PubMed and Embase layers complete (2005-2026). Conference material is retrieved and reported but excluded by the publication-form criterion.

Included studies per publication year



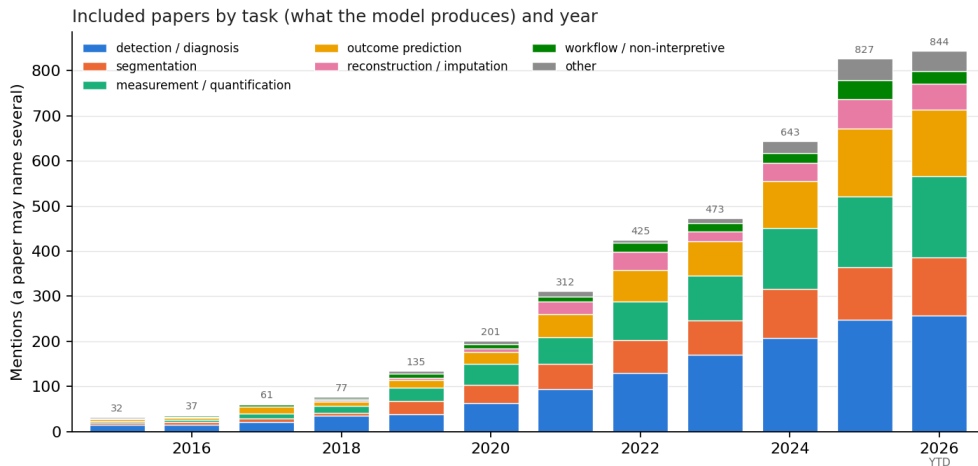
Blue = the 3,496 included primary studies; gray = records read and then excluded as off-topic or non-primary. 2025+ is year to date and still being indexed.

What the pediatric literature is made of

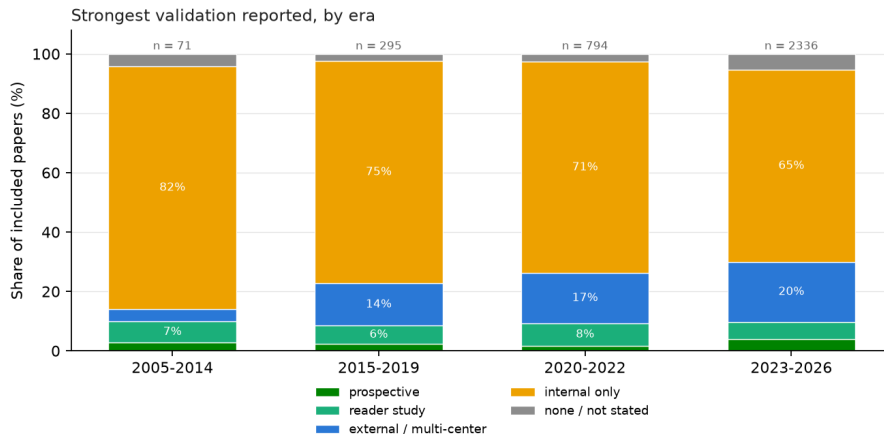


Modality: MRI 46%, ultrasound 24%, x-ray / radiography 19%, CT 9%. **Task:** detection / diagnosis 38%, measurement / quantification 24%, outcome prediction 20%, segmentation 19%. **Age group:** child 37%, adolescent 25%, fetal 22%, infant 15%. Categories are multilabel, so an axis can exceed 100%. MRI leads on the strength of the fetal and developmental-neuroscience literature: remove the MRI studies that use cognitive-neuroscience methods or report cognitive and behavioral outcomes and MRI falls level with ultrasound (30% each), which is why reviews built on narrower vocabulary rank radiography or ultrasound first.

What the models produce

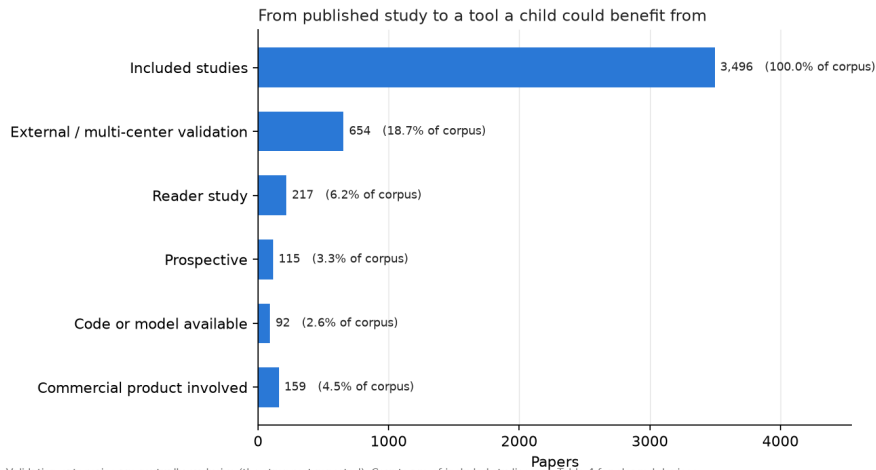


How well is it tested?



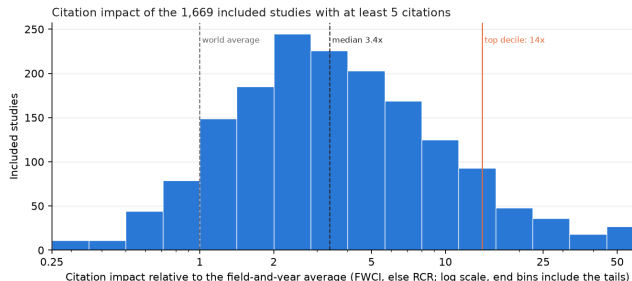
Of 3,496 included studies, external or multi-center validation was reported by 18.7%, a reader study by 6.2%, prospective evaluation by 3.3%, and a working code or model URL by 92 studies. External validation rose across eras; reader studies did not.

From published study to a tool a child benefits from



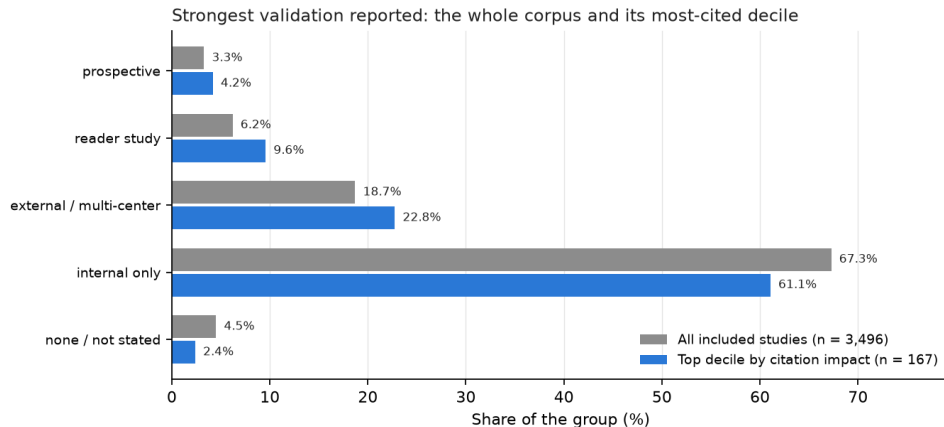
Validation categories are mutually exclusive (the strongest reported). Counts are of included studies; see Table 4 for cleared devices.

Which of these studies is the field actually reading?



- The median included study is cited 3.38 times the world average for its field and year: this is a well-cited literature, not a neglected one.
- 1,669 of 3,496 studies (47.7%) have enough citations to be ranked; the rest are too recent.
- The top decile is the 167 studies at 14x average and up (PubMed 158, Embase 6, OpenAlex 3).
- Selecting on citations does not select for rigor: external or multi-center validation is 22.8% in that decile against 18.7% across the corpus, and two thirds of it is still internal validation only.

Does citation impact pick out the better-validated work?



The most-cited decile is somewhat more likely to report external validation and a reader study, but the difference is small: citations track topic and audience more than evidence.

Highest-impact pediatric radiology AI studies, 2005–2026

Impact	Cites	Year	Study	Venue	What it does
329x	273	2018	Performance of a Deep-Learning Neural Network Model in Assessing Skeletal Maturity on Pediatric Hand Radiographs.	Radiology	skeletal maturity (bone age) assessment in the workup of endocrine and metabolic disorders
325x	219	2017	Fully Automated Deep Learning System for Bone Age Assessment.	Journal of digital imaging	skeletal maturity (bone age) assessment in endocrine and metabolic workup
293x	1,757	2018	Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning.	Cell	pediatric pneumonia on chest radiographs, and bacterial vs viral origin
221x	167	2017	Deep learning for automated skeletal bone age assessment in X-ray images.	Medical image analysis	skeletal maturity (bone age) assessment in endocrine, genetic and growth disorders
200x	321	2016	Automatic Segmentation of MR Brain Images With a Convolutional Neural Network.	IEEE transactions on medical imaging	automatic brain tissue segmentation for quantitative MRI analysis
93x	269	2021	CA-Net	IEEE transactions on medical imaging	segmenting the placenta and fetal brain on fetal MRI
83x	701	2019	Attention Gate (Attention U-Net)	Medical image analysis	identifying the correct standard views during fetal ultrasound screening
73x	112	2017	Computerized Bone Age Estimation Using Deep Learning Based Program: Evaluation of the Accuracy and Efficiency.	AJR. American journal of roentgenology	bone age estimation, and whether software assistance makes radiologists faster without making them worse
71x	78	2019	Artificial intelligence-assisted interpretation of bone age radiographs improves accuracy and decreases variability.	Skeletal radiology	whether AI assistance makes pediatric radiologists better at bone age, not whether AI alone beats them

Impact = citations relative to the average paper of the same field and year (OpenAlex FWCI; iCite RCR where OpenAlex has none), shown once a study has 5 citations. † Embase-only; * preprint.

Highest-impact pediatric radiology AI studies, 2023

Impact	Cites	Study	Venue	What it does
69x	106	iBEAT V2.0	Nature protocols	measuring cortical development in infants, whose brain MRI defeats adult-designed processing tools
37x	20	Bone Age Assessment Using Artificial Intelligence in Korean Pediatric Population: A Comparison of Deep-Learning Models Trained With Healthy Chronological and Greulich-Pyle Ages as Labels.	Korean journal of radiology	whether bone age in a given population should be predicted against the Greulich-Pyle atlas or learned directly from the chronological ages of healthy children in that population
34x	9	Bone age assessment based on deep neural networks with annotation-free cascaded critical bone region extraction.	Frontiers in artificial intelligence	automated bone age assessment from hand radiographs without manual region annotation
30x	39	FeTA	Medical image analysis	segmenting the developing fetal brain into tissues, the step every quantitative prenatal neurodevelopment study depends on
28x	20	Deep focus approach for accurate bone age estimation from lateral cephalogram.	Journal of dental sciences	estimating skeletal maturity from the lateral cephalogram a growing child already has, so that a separate hand-wrist radiograph is not required
27x	11	Bone Age Estimation and Prediction of Final Adult Height Using Deep Learning.	Yonsei medical journal	automatically estimating bone age and predicting final adult height in children using hand radiographs
26x	21	Detection of ASD Children through Deep-Learning Application of fMRI.	Children (Basel, Switzerland)	early detection of autism spectrum disorder in children, where prompt diagnosis enables targeted intervention
26x	14	High performance for bone age estimation with an artificial intelligence solution.	Diagnostic and interventional imaging	How does an AI solution for bone age assessment compare in accuracy to a general (non-pediatric-specialized) radiologist?
25x	49	Automated classification of common maternal fetal ultrasound planes using multi-layer perceptron with deep feature integration †	Biomedical Signal Processing and Control	automated identification of standard fetal ultrasound planes during obstetric examination

410 studies included from 2023; 289 have enough citations to be ranked. Impact = citations relative to the average paper of the same field and year (OpenAlex FWCI; iCite RCR where OpenAlex has none), shown once a study has 5 citations. † Embase-only; * preprint.

Highest-impact pediatric radiology AI studies, 2024

Impact	Cites	Study	Venue	What it does
29x	16	Automated bone age assessment in a German pediatric cohort: agreement between an artificial intelligence software and the manual Greulich and Pyle method.	European radiology	whether a bone age product can be swapped in for an expert reader, not merely correlate with one
29x	16	A Dempster-Shafer Approach to Trustworthy AI With Application to Fetal Brain MRI Segmentation.	IEEE transactions on pattern analysis and machine intelligence	detecting and correcting failures in automated fetal brain MRI segmentation so results are safe to use clinically
26x	12	HFSCCD: A Hybrid Neural Network for Fetal Standard Cardiac Cycle Detection in Ultrasound Videos.	IEEE journal of biomedical and health informatics	Automatically detecting standard cardiac cycle frames in fetal echocardiography to support congenital heart disease diagnosis
24x	5	Assessment of Bone Age Based on Hand Radiographs Using Regression-Based Multi-Modal Deep Learning.	Life (Basel, Switzerland)	estimating bone age from hand radiographs
24x	17	FetalBrainAwareNet	Computerized medical imaging and graphics	Synthesizing anatomically accurate fetal head standard-plane ultrasound images to augment training data for automated fetal head standard-plane classification.
24x	16	On the use of contrastive learning for standard-plane classification in fetal ultrasound imaging.	Computers in biology and medicine	automated identification of standard imaging planes in fetal ultrasound examinations
22x	12	Transformer based multi-modal MRI fusion for prediction of post-menstrual age and neonatal brain development analysis.	Medical image analysis	Estimating post-menstrual age and assessing brain maturity from neonatal brain MRI to help identify preterm-related developmental risk
21x	9	TW3 AI bone age assessment system	Quantitative imaging in medicine and surgery	validating an AI system for Tanner-Whitehouse 3 bone age assessment across multiple children's hospitals
20x	11	Artificial intelligence assisted common maternal fetal planes prediction from ultrasound images based on information fusion of customized convolutional neural networks.	Frontiers in medicine	automatically classifying standard maternal-fetal ultrasound planes/organs to aid interpretation

555 studies included from 2024; 264 have enough citations to be ranked.

Highest-impact pediatric radiology AI studies, 2025

Impact	Cites	Study	Venue	What it does
54x	10	ChatGPT-o4-mini-high, ChatGPT-4.5, Gemini 2.5 Pro	Academic radiology	whether a general-purpose chatbot can substitute for dedicated bone age software, which is expensive
52x	34	Prediction of mental health risk in adolescents.	Nature medicine	predicting future risk of general psychiatric illness in adolescents
32x	13	BoneXpert, PANDA, BoneView	European radiology	which of the bone age products approved for the European market a department should buy
29x	10	Whole examination AI estimation of fetal biometrics from 20-week ultrasound scans.	NPJ digital medicine	fetal biometry currently rests on a handful of frames a sonographer chose, and the choice is a source of error
29x	8	Uncovering ethical biases in publicly available fetal ultrasound datasets.	NPJ digital medicine	whether publicly available fetal ultrasound image datasets used to train deep-learning prenatal-diagnosis algorithms are demographically and clinically representative
28x	13	Scanner-based real-time three-dimensional brain + body slice-to-volume reconstruction for T2-weighted 0.55-T low-field fetal magnetic resonance imaging.	Pediatric radiology	motion-corrected 3D fetal MRI has stayed a research technique because reconstruction happens offline, after the patient has gone home
26x	21	Predicting pediatric patient rehabilitation outcomes after spinal deformity surgery with artificial intelligence.	Communications medicine	telling an adolescent before scoliosis surgery how they are likely to feel afterwards, which radiographic correction alone does not predict
24x	10	Three-dimensional markerless surface topography approach with convolutional neural networks for adolescent idiopathic scoliosis screening.	Scientific reports	screening adolescents for scoliosis without the unnecessary referrals and radiographs that current screening generates
23x	5	Advancing prenatal healthcare by explainable AI enhanced fetal ultrasound image segmentation using U-Net++ with attention mechanisms.	Scientific reports	automated, explainable segmentation of fetal ultrasound images to standardize evaluation of fetal development

685 studies included from 2025; 118 have enough citations to be ranked.

Highest-impact pediatric radiology AI studies, 2026 (year to date)

Impact	Cites	Study	Venue	What it does
46x	7	Infants have rich visual categories in ventrotemporal cortex at 2 months of age.	Nature neuroscience	characterizing the development of visual category representations in the infant brain using functional MRI, compared against a deep neural network model of vision
14x	6	Perceived Racism, Brain Development, and Internalizing and Externalizing Symptoms: Findings From the ABCD Study.	Journal of the American Academy of Child and Adolescent Psychiatry	assessing effects of perceived racial discrimination on longitudinal subcortical functional connectivity changes and internalizing/externalizing symptoms in adolescents
6x	5	Use of Artificial Intelligence-Based Software to Aid in the Identification of Ultrasound Findings Associated With Fetal Congenital Heart Defects.	Obstetrics and gynecology	evaluating whether AI-based software improves ob-gyn and maternal-fetal medicine specialists' identification of second-trimester fetal ultrasound findings suspicious for congenital heart defects
5x	5	Artificial Intelligence for the Detection of Fetal Ultrasound Findings Concerning for Major Congenital Heart Defects.	Obstetrics and gynecology	identifying second-trimester fetal ultrasound findings suspicious for major congenital heart defects using AI-based software, to improve antenatal CHD detection
4x	14	G-YOLOv11 †	Biomedical Signal Processing and Control	fracture detection on pediatric wrist radiographs
4x	8	Machine Learning Algorithms for Predicting Cobb Angle Beyond 25° in Female Adolescent Idiopathic Scoliosis Patients.	Spine	predicting which female adolescents with idiopathic scoliosis will progress to a Cobb angle beyond 25°, using minimal radiographic and questionnaire data collected at the first visit
3x	5	Motion-Echo	European heart journal	automating myocardial strain analysis from pediatric echocardiograms to standardize measurement and enable earlier detection of cardiac dysfunction
3x	8	Safe for kids? AI medical devices in radiology overlook paediatric suitability.	European radiology	whether the AI devices a European radiology department can buy say anything about being safe for children
3x	9	GPT-4.1	European archives of oto-rhino-laryngology	detecting and lateralizing pediatric airway foreign body aspiration from clinical data and chest radiographs

686 studies included from 2026; 12 have enough citations to be ranked. Few studies from this year have enough citations for a normalized value yet, so this table is thin by construction rather than because the year was quiet.

The biggest public pediatric radiology AI datasets

Dataset	Year	Modality	Size	Ages	Access
CBTN clinical MRI (Children's Brain Tumor Network)	2023	brain / spine tumor MRI	23,101 exams (1,526 patients); tumor masks for 370	pediatric	application (NCI CCDI)
GRAZPEDWRI-DX	2022	wrist radiographs (trauma)	20,327 images (6,091 patients)	0.2–19 y	open
RSNA Pediatric Bone Age Challenge	2017	hand radiographs	14,236 images (bone age + sex)	mean 10.6 y	open (RSNA terms)
FETAL_PLANES_DB	2020	fetal ultrasound (standard planes)	12,400 images (1,792 pregnancies)	fetal	open
ABCD Study	2018–	brain MRI, longitudinal	11,878 participants, up to 7 follow-ups	9–10 y at baseline	application (NBDC)
PediCXR / VinDr-PCXR	2023	chest radiographs	9,125 studies; 36 findings, 15 diagnoses	< 10 y	credentialed (PhysioNet)
Guangzhou pediatric chest X-ray (Kermany)	2018	chest radiographs	5,856 images (normal / bacterial / viral pneumonia)	1–5 y	open
ABIDE I + II (autism)	2012 / 2016	brain MRI (structural, resting fMRI)	2,156 subjects	5–64 y (mostly children)	open
Regensburg Pediatric Appendicitis	2023	abdominal ultrasound + labs	1,709 images (579 patients)	0–18 y	open
dHCP (Developing Human Connectome Project)	2022–24	neonatal + fetal brain MRI	886 neonatal + 297 fetal datasets, with segmentations	20–45 weeks	open (NDA)
Pediatric-CT-SEG (TCIA)	2022	chest / abdomen / pelvis CT	718 series (359 patients); 29 organ contours	5 days – 16 y	open
BraTS-PEDs (pediatric brain tumor challenge)	2023–25	brain tumor MRI	464 patients (2024 challenge); 1,649 series on TCIA	pediatric	registration / TCIA
FeTA (fetal brain tissue annotation)	2021–24	fetal brain MRI	280 volumes (120 train + 160 test); 7 tissue labels	18–35 weeks	registration

Sizes verified against the primary paper or hosting page (2026-09). Only two pediatric radiograph sets exceed 10,000 images; the largest public pediatric-only CT set has 359 patients; no RSNA pediatric challenge has run since bone age in 2017. Adult benchmarks (CheXpert 224k, MIMIC-CXR 377k) exclude children.

Most-cited radiology AI papers, 2008-2022

Cites	FWCI	Paper	Venue	What it answers
14,326	1023.0	A survey on deep learning in medical image analysis (2017)	Medical Image Anal.	<i>survey</i> : What can deep learning do across medical imaging? (the field's reference survey)
5,658	446.6	Swin-Unet: Unet-like Pure Transformer for Medical Image Segmentation (2021)	ECCV Workshops	<i>segmentation</i>
2,990	340.4	Deep Learning in Medical Image Analysis (2017)	Annual Review of Biomedical Engineering	<i>survey</i> : How does deep learning apply to medical image analysis? (review)
2,553	290.0	Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images (2022)	BrainLes@MICCAI	<i>segmentation</i>
2,769	272.4	COVID-Net: a tailored deep convolutional neural network design for detection of COVID-19 cases from chest X-ray images (2020)	Scientific Reports	<i>classification</i> : Can a CNN detect COVID-19 on chest X-ray? (COVID-Net)
3,634	236.9	ChestX-Ray8: Hospital-Scale Chest X-Ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases (2017)	Computer Vision and Pattern Recognition	<i>classification</i>
10,044	226.3	nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation (2020)	Nature Methods	<i>segmentation</i> : Can one self-configuring method segment any organ or lesion without hand tuning? (nnU-Net)
4,100	170.2	Convolutional neural networks: an overview and application in radiology (2018)	Insights into Imaging	<i>tutorial</i> : What is a convolutional neural network, explained for radiologists?
5,587	158.4	Computational Radiomics System to Decode the Radiographic Phenotype (2017)	Cancer Research	<i>radiomics</i> : Can standardized image features (radiomics) be extracted reproducibly? (PyRadiomics)
3,322	149.0	Artificial intelligence in radiology (2018)	Nature Reviews. Cancer	<i>review</i> : How will AI change radiology practice? (Hosny et al.)

Rows are the most-cited papers of the period, ordered by FWCI = field-weighted citation impact (OpenAlex): citations relative to the world average for papers of the same field and year (1 = average), so a recent paper can be compared with an older one. Preprints (arXiv, medRxiv) are included; the venue column names the server.

Most-cited radiology AI papers, 2023

Cites	FWCI	Paper	Venue	What it answers
1,608	615.8	Segment anything in medical images	Nature Communications	<i>foundation</i> : Can a prompt-based 'segment anything' model work on medical images? (MedSAM)
2,099	243.8	Foundation models for generalist medical artificial intelligence	Nature	<i>foundation model</i> : What would a single generalist medical AI model that handles images, text and records look like? (perspective)
702	118.5	Redefining Radiology: A Review of Artificial Intelligence Integration in Medical Imaging	Diagnostics	<i>survey / review</i> : Where does AI fit in radiology workflow, from acquisition to reporting? (review)
509	75.2	Medical image analysis using deep learning algorithms	Frontiers in Public Health	<i>other</i>
562	74.5	A Study of CNN and Transfer Learning in Medical Imaging: Advantages, Challenges, Future Scope	Sustainability	<i>other</i>
861	69.6	Segment Anything Model for medical image analysis: an experimental study	Medical Image Anal.	<i>segmentation</i> : How well does the unmodified Segment Anything Model work on medical images? (it needs prompts and adaptation)
573	55.8	MRI-based brain tumor detection using convolutional deep learning methods and chosen machine learning techniques	BMC Medical Informatics and Decision Making	<i>oncology</i>
460	44.5	Brain Tumor Detection Based on Deep Learning Approaches and Magnetic Resonance Imaging	Cancers	<i>oncology</i>
391	42.2	SAM-Adapter: Adapting Segment Anything in Underperformed Scenes	2023 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)	<i>segmentation</i>
442	37.4	A generalist vision-language foundation model for diverse biomedical tasks	Nature Medicine	<i>other</i>

Rows are the most-cited papers of the period, ordered by FWCI = field-weighted citation impact (OpenAlex): citations relative to the world average for papers of the same field and year (1 = average), so a recent paper can be compared with an older one. Preprints (arXiv, medRxiv) are included; the venue column names the server. 1 of the 10 rows are preprints.

Most-cited radiology AI papers, 2024

Cites	FWCI	Paper	Venue	What it answers
1,074	225.2	VM-UNet: Vision Mamba UNet for Medical Image Segmentation	ACM Transactions on Multimedia Computing, Communications, and Applications (TOMCCAP)	<i>segmentation</i> : Does a state-space (Mamba) backbone beat convolutions and transformers for medical segmentation?
242	110.3	Generalist foundation models from a multimodal dataset for 3D computed tomography	Nature Biomedical Engineering	<i>other</i>
282	74.6	Artificial intelligence in neuro-oncology: advances and challenges in brain tumor diagnosis, prognosis, and precision treatment	npj Precision Oncology	<i>oncology</i>
279	72.6	METHodological RadiomIcs Score (METRICS): a quality scoring tool for radiomics research endorsed by EuSoMI	Insights into Imaging	<i>radiomics</i> : How should the quality of a radiomics study be scored? (METRICS checklist)
349	47.6	Deep learning for medical image segmentation: State-of-the-art advancements and challenges	Informatics in Medicine Unlocked	<i>segmentation</i>
317	43.4	Advances in Medical Image Segmentation: A Comprehensive Review of Traditional, Deep Learning and Hybrid Approaches	Bioengineering	<i>segmentation</i>
238	37.0	Employing deep learning and transfer learning for accurate brain tumor detection	Scientific Reports	<i>oncology</i>
242	36.0	Deep semi-supervised learning for medical image segmentation: A review	Expert systems with applications	<i>segmentation</i>
241	34.5	A review of deep learning-based information fusion techniques for multimodal medical image classification	Comput. Biol. Medicine	<i>survey / review</i>
298	31.8	Vision-language models for medical report generation and visual question answering: a review	Frontiers Artif. Intell.	<i>report generation / LLM</i> : What can vision-language models do for report generation and visual question answering? (review)

Most-cited radiology AI papers, 2025

Cites	FWCI	Paper	Venue	What it answers
258	225.8	Deep Convolutional Neural Networks in Medical Image Analysis: A Review	Inf.	<i>survey / review</i>
145	158.8	Multi-model assurance analysis showing large language models are highly vulnerable to adversarial hallucination attacks during clinical decision support	Communications Medicine	<i>LLM / reports</i>
122	108.2	Multimodal generative AI for medical image interpretation	Nature	<i>other</i>
313	94.9	Towards generalist foundation model for radiology by leveraging web-scale 2D&3D medical data	Nature Communications	<i>foundation model</i> : Can a radiology foundation model trained on web-scale 2D and 3D scans generalize across modalities and tasks? (RadFM)
173	87.2	Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology	Nature Cancer	<i>agent / autonomous</i> : Can an autonomous LLM agent that calls imaging and other tools support oncology decisions? (validation study)
208	65.9	A review of deep learning for brain tumor analysis in MRI	npj Precision Oncology	<i>survey / review</i>
150	65.8	Med-R1: Reinforcement Learning for Generalizable Medical Reasoning in Vision-Language Models	IEEE Transactions on Medical Imaging	<i>LLM / reports</i>
233	64.6	Artificial intelligence in healthcare and medicine: clinical applications, therapeutic advances, and future perspectives	European Journal of Medical Research	<i>other</i>
151	54.0	Medical Image Segmentation: A Comprehensive Review of Deep Learning-Based Methods	Tomography	<i>segmentation</i>
118	39.5	Advanced Brain Tumor Classification in MR Images Using Transfer Learning and Pre-Trained Deep CNN Models	Cancers	<i>oncology</i>

1 of the 10 rows are preprints.

Most-cited radiology AI papers, 2026 (year to date)

Cites	FWCI	Paper	Venue	What it answers
48	102.6	A generalizable foundation model for analysis of human brain MRI	Nature Neuroscience	<i>other</i>
16	90.5	Reporting checklist for foundation and large language models in medical research (REFINE): an international consensus guideline.	Diagnostic and Interventional Radiology	<i>LLM / reports</i>
15	76.1	Scaling medical AI across clinical contexts	Nature Medicine	<i>foundation model</i> : How can medical AI be scaled across hospitals, populations and tasks without breaking? (perspective)
27	67.3	Applications and limitations of large language models to integrate medical context: a comprehensive review	Iran Journal of Computer Science	<i>survey / review</i>
18	62.7	A hybrid deep learning framework based on VGG19 and U-Net for accurate brain tumor segmentation in MRI images	Magnetic Resonance Letters	<i>segmentation</i>
37	58.8	Explainable artificial intelligence (XAI) in medical imaging: a systematic review of techniques, applications, and challenges	BMC Medical Imaging	<i>survey / review</i> : Which explainability methods are used in medical imaging AI and do they help? (review)
12	45.3	Explainable AI-driven MRI-based brain tumor classification: a novel deep learning approach	Frontiers Artif. Intell.	<i>oncology</i>
21	44.2	Agentic AI in Radiology: Evolution from Large Language Models to Future Clinical Integration.	Radiology: Artificial Intelligence	<i>agent / autonomous</i> : What are agentic AI systems in radiology and what stands between them and clinical use? (review)
17	28.4	Leveraging multi-modal foundation models for analysing spatial multi-omic and histopathology data	Nature Biomedical Engineering	<i>other</i>
19	21.4	BIBSNet: A deep learning baby image brain segmentation network for MRI scans	Developmental Cognitive Neuroscience	<i>segmentation</i> : Can one network segment infant brain MRI across the first years of life? (BIBSNet)

Citations for the current year are still accumulating; the FWCI column is the more stable signal here. 1 of the 10 rows are preprints.

Most-cited pediatric radiology AI papers, 2008-2022

Cites	FWCI	Paper	Venue	What it answers
426	328.6	Performance of a Deep-Learning Neural Network Model in Assessing Skeletal Maturity on Pediatric Hand Radiographs. (2017)	Radiology	<i>bone age</i> : Can a CNN assess skeletal maturity as accurately as radiologists?
431	321.5	Fully Automated Deep Learning System for Bone Age Assessment (2017)	Journal of digital imaging	<i>bone age</i> : Fully automated bone age from hand radiographs
403	221.2	Deep learning for automated skeletal bone age assessment in X-ray images (2017)	Medical Image Anal.	<i>bone age</i> : Deep learning for automated bone age (BoNet)
143	73.0	Computerized Bone Age Estimation Using Deep Learning Based Program: Evaluation of the Accuracy and Efficiency. (2017)	AJR. American journal of roentgenology	<i>bone age</i> : Is a commercial deep-learning bone-age program accurate and faster in practice?
191	57.8	A Review on Deep-Learning Algorithms for Fetal Ultrasound-Image Analysis (2022)	Medical Image Anal.	<i>survey / review</i>
352	57.4	Standard Plane Localization in Fetal Ultrasound via Domain Transferred Deep Neural Networks (2015)	IEEE journal of biomedical and health informatics	<i>fetal MRI</i>
194	57.0	Automatic Fetal Ultrasound Standard Plane Recognition Based on Deep Learning and IIoT (2021)	IEEE Transactions on Industrial Informatics	<i>fetal MRI</i>
233	30.3	Evaluation of deep convolutional neural networks for automatic classification of common maternal fetal ultrasound planes (2020)	Scientific Reports	<i>fetal MRI</i>
458	27.0	The RSNA Pediatric Bone Age Machine Learning Challenge. (2019)	Radiology	<i>bone age</i> : How well can crowdsourced AI estimate bone age? (RSNA challenge, 260 teams)
253	25.8	An ensemble of neural networks provides expert-level prenatal detection of complex congenital heart disease (2021)	Nature Medicine	<i>detection</i>

A different corpus from the systematic-review tables earlier in the deck: this is a citation search of Semantic Scholar over radiology AI with a pediatric title filter, so it includes conference papers and is not screened for eligibility. The review tables rank the screened corpus. Both are shown because neither contains the other.

Most-cited pediatric radiology AI papers, 2023

Cites	FWCI	Paper	Venue	What it answers
32	36.6	Bone Age Assessment Using Artificial Intelligence in Korean Pediatric Population: A Comparison of Deep-Learning Models Trained With Healthy Chronological and Greulich-Pyle Ages as Labels	Korean Journal of Radiology	<i>bone age</i>
62	33.6	Standard fetal ultrasound plane classification based on stacked ensemble of deep learning models	Expert systems with applications	<i>fetal ultrasound</i> : Can an ensemble of networks classify the standard fetal ultrasound planes?
27	33.5	Bone age assessment based on deep neural networks with annotation-free cascaded critical bone region extraction	Frontiers in Artificial Intelligence	<i>bone age</i>
43	23.8	Review on deep learning fetal brain segmentation from Magnetic Resonance images	Artif. Intell. Medicine	<i>segmentation</i>
37	20.2	Transfer learning for accurate fetal organ classification from ultrasound images: a potential tool for maternal healthcare providers	Scientific Reports	<i>fetal MRI</i>
34	16.6	Deep Learning-Based Multiclass Brain Tissue Segmentation in Fetal MRIs	Italian National Conference on Sensors	<i>segmentation</i>
43	9.7	Deeplasia: deep learning for bone age assessment validated on skeletal dysplasias	Pediatric Radiology	<i>bone age</i>
30	7.8	Deep learning for estimation of fetal weight throughout the pregnancy from fetal abdominal ultrasound.	American Journal of Obstetrics & Gynecology MFM	<i>fetal MRI</i>
38	7.3	Deep learning model for prenatal congenital heart disease (CHD) screening can be applied to retrospective imaging from the community setting, outperforming initial clinical detection in a well-annotated cohort	Ultrasound in Obstetrics and Gynecology	<i>fetal ultrasound</i> : Does a prenatal congenital heart disease screening model work on retrospective real-world scans?
31	5.8	MRI-Based End-To-End Pediatric Low-Grade Glioma Segmentation and Classification	Canadian Association of Radiologists journal = Journal l'Association canadienne des radiologistes	<i>segmentation</i>

Rows are the most-cited papers of the period, ordered by FWCI = field-weighted citation impact (OpenAlex): citations relative to the world average for papers of the same field and year (1 = average), so a recent paper can be compared with an older one. Preprints (arXiv, medRxiv) are included; the venue column names the server. 1 of the 10 rows are preprints.

Most-cited pediatric radiology AI papers, 2024

Cites	FWCI	Paper	Venue	What it answers
17	34.8	Intrapartum Ultrasound Image Segmentation of Pubic Symphysis and Fetal Head Using Dual Student-Teacher Framework with CNN-VIT Collaborative Learning	International Conference on Medical Image Computing and Computer-Assisted Intervention	<i>segmentation</i>
30	23.8	FetalBrainAwareNet: Bridging GANs with anatomical insight for fetal ultrasound brain plane synthesis	Comput. Medical Imaging Graph.	<i>fetal MRI</i>
20	19.5	Artificial intelligence assisted common maternal fetal planes prediction from ultrasound images based on information fusion of customized convolutional neural networks	Frontiers in Medicine	<i>fetal MRI</i>
18	15.9	Deep learning denoising reconstruction for improved image quality in fetal cardiac cine MRI	Frontiers in Cardiovascular Medicine	<i>reconstruction</i> : Does deep-learning denoising improve fetal cardiac cine MRI?
30	15.1	RTSeg-net: A lightweight network for real-time segmentation of fetal head and pubic symphysis from intrapartum ultrasound images	Comput. Biol. Medicine	<i>segmentation</i>
16	13.4	FetSAM: Advanced Segmentation Techniques for Fetal Head Biometrics in Ultrasound Imagery	IEEE Open Journal of Engineering in Medicine and Biology	<i>segmentation</i>
23	7.0	Deep learning-based automated lesion segmentation on pediatric focal cortical dysplasia II preoperative MRI: a reliable approach	Insights into Imaging	<i>epilepsy</i> : Can focal cortical dysplasia lesions be found automatically on preoperative pediatric MRI?
18	5.3	Integration of a Deep Convolutional Neural Network With Adaptive Channel Weight Technique for Automated Identification of Standard Fetal Biometry Planes	IEEE Transactions on Instrumentation and Measurement	<i>fetal MRI</i>
18	5.2	Explainable AI in Deep Learning Based Classification of Fetal Ultrasound Image Planes	Procedia Computer Science	<i>fetal MRI</i>
19	4.8	A deep learning framework for identifying and segmenting three vessels in fetal heart ultrasound images	BioMedical Engineering OnLine	<i>segmentation</i>

Most-cited pediatric radiology AI papers, 2025

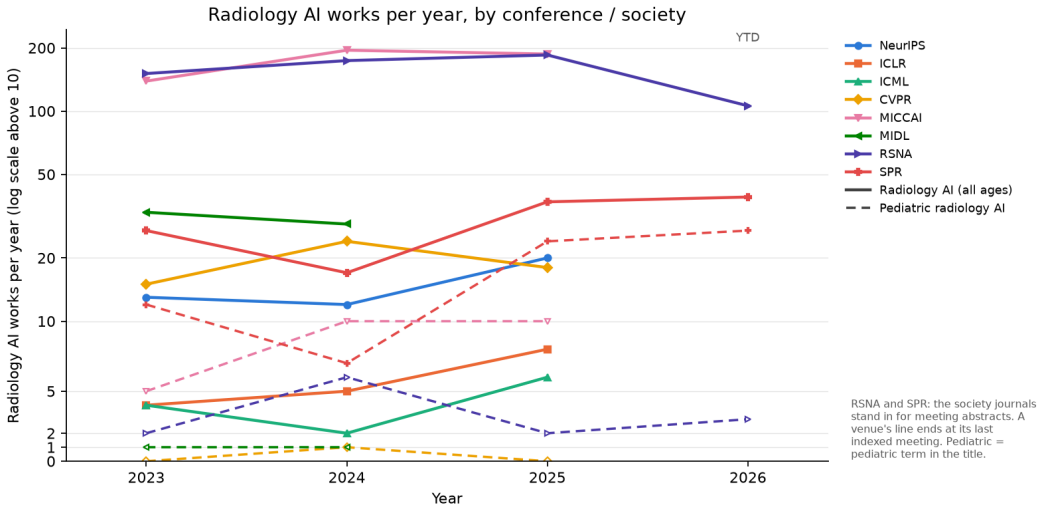
Cites	FWCI	Paper	Venue	What it answers
16	171.3	A comprehensive review of artificial intelligence - based algorithm towards fetal facial anomalies detection (2013–2024)	Artificial Intelligence Review	<i>survey / review</i>
18	85.1	M-CNN-RF: A hybrid deep learning model for accurate pediatric skeletal age estimation using hand bone radiographs	Alexandria Engineering Journal	<i>other</i>
20	77.0	Bone Age Assessment Using Various Medical Imaging Techniques Enhanced by Artificial Intelligence	Diagnostics	<i>bone age</i>
18	24.9	Advancing prenatal healthcare by explainable AI enhanced fetal ultrasound image segmentation using U-Net++ with attention mechanisms	Scientific Reports	<i>segmentation</i>
12	21.8	FetalMovNet: A Novel Deep Learning Model Based on Attention Mechanism for Fetal Movement Classification in US	IEEE Access	<i>fetal MRI</i>
11	14.5	Effectiveness and clinical impact of using deep learning for first-trimester fetal ultrasound image quality auditing	BMC Pregnancy and Childbirth	<i>fetal MRI</i>
12	12.4	Association between deep learning radiomics based on placental MRI and preeclampsia with fetal growth restriction: A multicenter study.	European Journal of Radiology	<i>radiomics</i>
16	8.7	Prenatal detection of congenital heart defects using the deep learning-based image and video analysis: protocol for Clinical Artificial Intelligence in Fetal Echocardiography (CAIFE), an international multicentre multidisciplinary study	BMJ Open	<i>fetal ultrasound: Can deep learning on prenatal ultrasound video detect congenital heart defects?</i>
11	8.6	Multi-view deep learning framework for the detection of chest X-rays compatible with pediatric pulmonary tuberculosis	Nature Communications	<i>detection</i>
11	7.0	Evaluation of Image Quality and Scan Time Efficiency in Accelerated 3D T1-Weighted Pediatric Brain MRI Using Deep Learning-Based Reconstruction	Korean Journal of Radiology	<i>reconstruction</i>

Most-cited pediatric radiology AI papers, 2026 (year to date)

Cites	FWCI	Paper	Venue	What it answers
11	34.3	Artificial intelligence and radiomics for pediatric brain tumor classification and molecular characterization: a systematic review	Neuroradiology	<i>oncology</i> : Can AI and radiomics classify pediatric brain tumors and their molecular subtypes? (review)
5	27.9	Deep learning assessment of fetal brain maturation on 3D ultrasound volumes in early-onset fetal growth restriction	Ultrasound in Obstetrics and Gynecology	<i>fetal ultrasound</i> : Can deep learning grade fetal brain maturation on 3D ultrasound in growth restriction?
3	17.6	Foundation models in radiology: a primer for pediatric radiologists	Pediatric Radiology	<i>foundation model</i> : What should a pediatric radiologist know about foundation models? (primer)
2	17.6	Deep Learning-Based Multi-Class Pediatric Wrist Fracture Subtype Classification: A Pilot Study Comparing Convolutional Neural Network Architectures	Journal of Imaging	<i>fracture</i>
3	17.3	Current applications and challenges of artificial intelligence applied to diagnostics in pediatric musculoskeletal imaging	Pediatric Radiology	<i>classification</i>
4	13.6	Data mining in pediatric radiology in the era of artificial intelligence.	Pediatric Radiology	<i>other</i>
2	13.2	Acquisition time/dose reduction in pediatric PET imaging using patch-based deep learning	EJNMMI Physics	<i>reconstruction</i> : Can pediatric PET use one fifth of the counts with patch-based deep-learning denoising?
2	12.6	Optimizing pediatric chest CT: superior image quality and lower radiation dose with deep learning reconstruction.	European Journal of Radiology	<i>reconstruction</i> : Can deep-learning reconstruction lower pediatric chest CT dose while improving image quality?
2	12.2	Enhancing efficiency in pediatric brain tumor segmentation using a pathologically diverse single-center clinical dataset	Neuro-Oncology Advances	<i>segmentation</i>
3	11.7	Automated bone age assessment in rare pediatric growth disorders: a comparative study using Deeplasia	Frontiers in Endocrinology	<i>bone age</i> : Does an open bone-age model hold up in rare growth disorders against several experts? (Deeplasia)

Citations for the current year are still accumulating; the FWCI column is the more stable signal here.

Radiology AI works per year, by conference



NeurIPS: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2025	24	–	3D-RAD: A Comprehensive 3D Radiology Med-VQA Dataset with Multi-Temporal Analysis and Diverse Diagnostic Tasks
2025	20	–	Better Tokens for Better 3D: Advancing Vision-Language Modeling in 3D Medical Imaging
2024	47	2.0	Touchstone Benchmark: Are We on the Right Way for Evaluating AI Algorithms for Medical Segmentation?
2024	26	0.9	Eye-gaze Guided Multi-modal Alignment for Medical Representation Learning
2024	25	1.2	Enhancing vision-language models for medical imaging: bridging the 3D gap with innovative slice selection
2024	20	–	DiffusionBlend: Learning 3D Image Prior through Position-aware Diffusion Score Blending for 3D Computed Tomography Reconstruction
2023	144	–	SegVol: Universal and Interactive Volumetric Medical Image Segmentation
2023	104	–	LVM-Med: Learning Large-Scale Self-Supervised Vision Models for Medical Imaging via Second-order Graph Matching
2023	72	–	EHRXQA: A Multi-Modal Question Answering Dataset for Electronic Health Records with Chest X-ray Images
2023	66	–	SwiFT: Swin 4D fMRI Transformer
2023	64	–	NeuroGraph: Benchmarks for Graph Machine Learning in Brain Connectomics
2023	19	–	Knowledge-based in silico models and dataset for the comparative evaluation of mammography AI for a range of breast characteristics, lesion conspicuities and doses

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 45 works, newest year first.

ICLR: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2025	70	–	Large-scale and Fine-grained Vision-language Pre-training for Enhanced CT Image Understanding
2025	13	–	SimXRD-4M: Big Simulated X-ray Diffraction Data and Crystal Symmetry Classification Benchmark
2025	9	–	Self-Supervised Diffusion MRI Denoising via Iterative and Stable Refinement
2025	9	–	Score-based Self-supervised MRI Denoising
2024	143	–	MMed-RAG: Versatile Multimodal RAG System for Medical Vision Language Models
2024	31	–	MediConfusion: Can you trust your AI radiologist? Probing the reliability of multimodal medical foundation models
2024	19	–	The Effect of Intrinsic Dataset Properties on Generalization: Unraveling Learning Differences Between Natural and Medical Images
2024	16	–	LeFusion: Controllable Pathology Synthesis via Lesion-Focused Diffusion Models
2023	107	–	Advancing Radiograph Representation Learning with Masked Record Modeling
2023	84	–	DDM2: Self-Supervised Diffusion MRI Denoising with Generative Diffusion Models
2023	40	–	AE-FLOW: Autoencoders with Normalizing Flows for Medical Images Anomaly Detection
2023	18	–	Improved Training of Physics-Informed Neural Networks Using Energy-Based Priors: a Study on Electrical Impedance Tomography

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 17 works, newest year first.

ICML: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2025	21	–	CLIMB: Data Foundations for Large Scale Multimodal Clinical Foundation Models
2025	17	–	MindLLM: A Subject-Agnostic and Versatile Model for fMRI-to-Text Decoding
2025	11	–	Raptor: Scalable Train-Free Embeddings for 3D Medical Volumes Leveraging Pretrained 2D Foundation Models
2025	5	–	Efficient Noise Calculation in Deep Learning-based MRI Reconstructions
2025	2	–	LangDAug: Langevin Data Augmentation for Multi-Source Domain Generalization in Medical Image Segmentation
2025	0	–	Staged and Physics-Grounded Learning Framework with Hyperintensity Prior for Pre-Contrast MRI Synthesis
2024	14	–	Unsupervised Domain Adaptation for Anatomical Structure Detection in Ultrasound Images
2024	10	–	Adaptive Sampling of k-Space in Magnetic Resonance for Rapid Pathology Prediction
2023	17	–	ACAT: Adversarial Counterfactual Attention for Classification and Detection in Medical Imaging
2023	16	–	Robustness of Deep Learning for Accelerated MRI: Benefits of Diverse Training Data
2023	15	–	Learning High-Order Relationships of Brain Regions
2023	8	–	Contextualized Policy Recovery: Modeling and Interpreting Medical Decisions with Adaptive Imitation Learning

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 12 works, newest year first.

Year	Cites	FWCI	Paper
2025	68	265.4	Enhanced Contrastive Learning with Multi-view Longitudinal Data for Chest X-ray Report Generation
2024	84	47.9	Towards Generalizable Tumor Synthesis
2024	81	14.6	Rethinking Diffusion Model for Multi-Contrast MRI Super-Resolution
2024	73	13.8	MemSAM: Taming Segment Anything Model for Echocardiography Video Segmentation
2023	243	20.2	Dynamic Graph Enhanced Contrastive Learning for Chest X-Ray Report Generation
2023	222	75.5	Ambiguous Medical Image Segmentation Using Diffusion Models
2023	217	18.7	METransformer: Radiology Report Generation by Transformer with Multiple Learnable Expert Tokens
2023	164	21.9	MCF: Mutual Correction Framework for Semi-Supervised Medical Image Segmentation
2023	133	14.4	KiUT: Knowledge-injected U-Transformer for Radiology Report Generation
2023	117	67.2	Structure-Aware Sparse-View X-Ray 3D Reconstruction
2023	103	11.0	Label-Free Liver Tumor Segmentation
2023	75	14.1	Learning Federated Visual Prompt in Null Space for MRI Reconstruction

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 57 works, newest year first.

MICCAI: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2025	208	1.1	MedVLM-R1: Incentivizing Medical Reasoning Capability of Vision-Language Models (VLMs) via Reinforcement Learning
2024	133	–	CT2Rep: Automated Radiology Report Generation for 3D Medical Imaging
2024	102	–	Anatomically-Controllable Medical Image Generation with Segmentation-Guided Diffusion Models
2024	87	–	MedCLIP-SAM: Bridging Text and Image Towards Universal Medical Image Segmentation
2023	462	–	MedNeXt: Transformer-driven Scaling of ConvNets for Medical Image Segmentation
2023	128	–	CoLa-Diff: Conditional Latent Diffusion Model for Multi-Modal MRI Synthesis
2023	105	–	DiffMIC: Dual-Guidance Diffusion Network for Medical Image Classification
2023	97	13.9	Beyond Adapting SAM: Towards End-to-End Ultrasound Image Segmentation via Auto Prompting
2023	95	–	Self-Supervised MRI Reconstruction with Unrolled Diffusion Models
2023	83	–	Make-A-Volume: Leveraging Latent Diffusion Models for Cross-Modality 3D Brain MRI Synthesis
2023	72	19.0	BerDiff: Conditional Bernoulli Diffusion Model for Medical Image Segmentation
2023	71	–	Ariadne's Thread: Using Text Prompts to Improve Segmentation of Infected Areas from Chest X-ray images

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 521 works, newest year first.

MIDL: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2024	14	–	Imbalance-aware loss functions improve medical image classification
2023	78	–	Patched Diffusion Models for Unsupervised Anomaly Detection in Brain MRI
2023	39	–	MMCFormer: Missing Modality Compensation Transformer for Brain Tumor Segmentation
2023	29	–	Medical diffusion on a budget: textual inversion for medical image generation
2023	23	–	MProtoNet: A Case-Based Interpretable Model for Brain Tumor Classification with 3D Multi-parametric Magnetic Resonance Imaging
2023	22	–	Diffusion Models for Contrast Harmonization of Magnetic Resonance Images
2023	20	–	Exploring Image Augmentations for Siamese Representation Learning with Chest X-Rays
2023	19	–	3D Medical Axial Transformer: A Lightweight Transformer Model for 3D Brain Tumor Segmentation
2023	19	–	Vision-Language Modelling For Radiological Imaging and Reports In The Low Data Regime
2023	14	–	$E(3) \times SO(3)$ -Equivariant Networks for Spherical Deconvolution in Diffusion MRI
2023	14	–	Amortized Normalizing Flows for Transcranial Ultrasound with Uncertainty Quantification
2023	13	–	Selective experience replay compression using coresets for lifelong deep reinforcement learning in medical imaging

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 62 works, newest year first.

RSNA: radiology AI works, 2023–2026

Year	Cites	FWCI	Paper
2024	452	113.7	Checklist for Artificial Intelligence in Medical Imaging (CLAIM): 2024 Update.
2024	324	26.4	Chatbots and Large Language Models in Radiology: A Practical Primer for Clinical and Research Applications.
2024	215	58.7	The Image Biomarker Standardization Initiative: Standardized Convolutional Filters for Reproducible Radiomics and Enhanced Clinical Insights.
2023	919	29.4	ChatGPT and Other Large Language Models Are Double-edged Swords.
2023	419	13.5	Performance of ChatGPT on a Radiology Board-style Examination: Insights into Current Strengths and Limitations.
2023	392	33.2	Deep Learning Image Reconstruction for CT: Technical Principles and Clinical Prospects.
2023	282	9.1	Automation Bias in Mammography: The Impact of Artificial Intelligence BI-RADS Suggestions on Reader Performance.
2023	273	44.4	Potential of ChatGPT and GPT-4 for Data Mining of Free-Text CT Reports on Lung Cancer.
2023	265	37.0	MRI-based Quantification of Intratumoral Heterogeneity for Predicting Treatment Response to Neoadjuvant Chemotherapy in Breast Cancer.
2023	259	47.2	How AI Responds to Common Lung Cancer Questions: ChatGPT vs Google Bard.
2023	258	35.5	A Guide to Cross-Validation for Artificial Intelligence in Medical Imaging.
2023	225	36.2	Predicting Microvascular Invasion in Hepatocellular Carcinoma Using CT-based Radiomics Model.

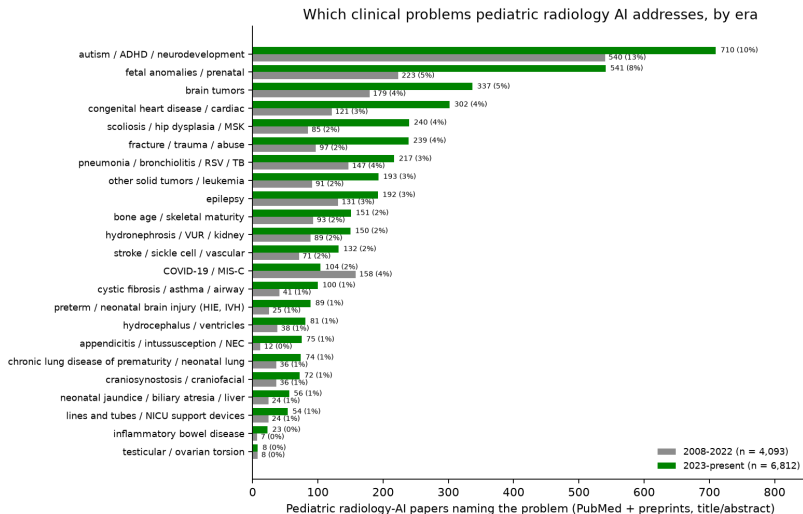
Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 616 works, newest year first.

SPR: radiology AI works, 2023–2026

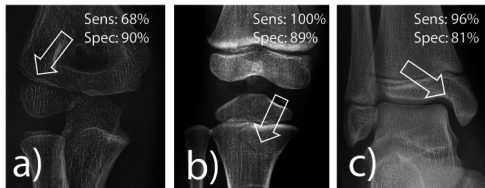
Year	Cites	FWCI	Paper
2025	18	4.6	AI implementation in pediatric radiology for patient safety: a multi-society statement from the ACR, ESPR, SPR, SLARP, AOSPR, SPIN *
2025	17	27.9	Scanner-based real-time three-dimensional brain + body slice-to-volume reconstruction for T2-weighted 0.55-T low-field fetal magnetic resonance imaging *
2024	24	2.0	Capability of multimodal large language models to interpret pediatric radiological images *
2024	15	–	Artificial Intelligence vs. Doctors: Diagnosing Necrotizing Enterocolitis on Abdominal Radiographs
2023	49	1.6	Detecting pediatric wrist fractures using deep-learning-based object detection *
2023	44	1.2	Artificial intelligence to identify fractures on pediatric and young adult upper extremity radiographs *
2023	43	9.7	Deeplasia: deep learning for bone age assessment validated on skeletal dysplasias *
2023	42	1.2	Comparison of diagnostic performance of a deep learning algorithm, emergency physicians, junior radiologists and senior radiologists in the detection of appendicular fractures in children
2023	35	1.0	Artificial intelligence-based detection of paediatric appendicular skeletal fractures: performance and limitations for common fracture types and locations *
2023	31	1.0	The unintended consequences of artificial intelligence in paediatric radiology *
2023	23	4.0	Artificial Intelligence in Paediatric Tuberculosis *
2023	20	4.2	Innovative advances in pediatric radiology: computed tomography reconstruction techniques, photon-counting detector computed tomography, and beyond *

Source: Semantic Scholar venue search, citations and FWCI from OpenAlex; title-level relevance filter; the 12 most-cited of 120 works, newest year first.

Which medical problems pediatric radiology AI addresses



AI fracture detection in a real pediatric emergency department

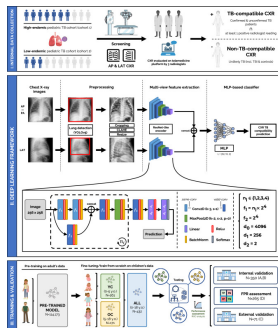


Ziegner et al., Eur Radiol 2025, figure 4: pediatric fractures the software found, with its per-region sensitivity and specificity

source: cdn.ncbi.nlm.nih.gov

- **Problem:** fractures in children are missed most often by the least experienced readers, at night, in the emergency department.
- **Data:** 1,672 consecutive radiographs of children under 18 from one tertiary pediatric emergency department, plus a selected medicolegal set (proximal tibia, radial condyle); three pediatric residents read each before and after seeing the AI.
- **Method:** a commercially available, CE-marked deep-learning fracture detector (adult-derived, pediatric extension) run stand-alone and as a second reader.
- **Result:** stand-alone sensitivity 92%, specificity 83%; 100% for proximal tibia but 68% for radial condyle fractures; residents' accuracy rose from 88% to 90%, and in 2% of cases they dropped a correct diagnosis after seeing the AI.
- **Why it matters:** the first real-life pediatric ED evaluation: the gain from a cleared adult tool is real but modest, and the elbow remains the weak spot.

pTBLightNet: pediatric tuberculosis on frontal and lateral chest radiographs



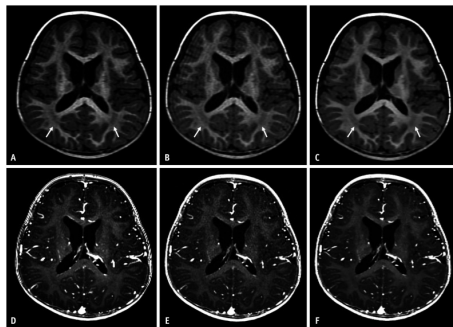
Capellan-Martin et al., Nat Commun 2025, figure 1: the multi-view

pTBLightNet pipeline

source: cdn.ncbi.nlm.nih.gov

- **Problem:** pediatric pulmonary TB has vaguer symptoms and subtler radiographs than adult TB; most affected children in high-burden settings are never diagnosed.
- **Data:** pre-trained on 114,173 adult chest radiographs, then fine-tuned and tested on 918 radiographs from three pediatric TB cohorts, with lateral views.
- **Method:** a multi-view network that scores frontal and lateral films for TB-compatible findings; separate models for under-5 and 5-18 years.
- **Result:** AUC 0.90 on internal and 0.68 on external testing; lateral views helped most in the youngest children; age-specific models matched larger undifferentiated ones.
- **Why it matters:** shows both the value of adult pre-training for a rare pediatric target and the internal-to-external drop that every pediatric deployment should expect.

Deep-learning reconstruction shortens pediatric brain MRI



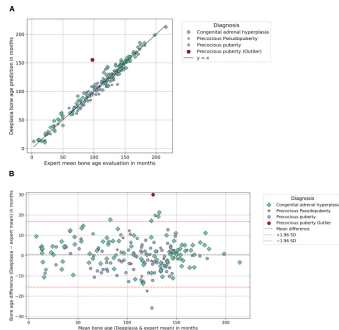
Yoo et al., Korean J Radiol 2025, figure 2: conventional (A, D) vs accelerated 3D T1 pediatric brain MRI without (B, E) and with (C, F)

deep-learning reconstruction

source: cdn.ncbi.nlm.nih.gov

- **Problem:** long 3D T1 sequences are where children move, sedation is prescribed, and scanner time is lost.
- **Data:** 46 children scanned with both the conventional and an accelerated 3D T1 protocol (pre- and post-contrast) on one 3T scanner.
- **Method:** vendor deep-learning reconstruction applied to the accelerated acquisition; noise, contrast, signal-to-noise and radiologist ratings compared with the conventional scan.
- **Result:** acquisition time cut by 29% (pre-contrast) and 41% (post-contrast); the deep-learning images had lower noise, higher SNR and CNR, better overall quality and fewer artifacts than the conventional scan.
- **Why it matters:** reconstruction AI is on the scanner today and gives children a shorter, better exam without any diagnostic model in the loop.

Deeplasia bone age in rare growth disorders, against multiple experts

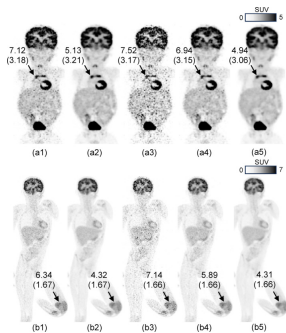


Skaf et al., Front Endocrinol 2026, figure 1: Deeplasia bone age vs the expert mean in the endocrine cohort (scatter and Bland-Altman)

source: cdn.ncbi.nlm.nih.gov

- **Problem:** bone-age models are trained on ordinary hands; the children who need bone age most have syndromes, endocrine disease or lysosomal storage disorders that distort the skeleton.
- **Data:** 1,138 hand radiographs from several centers: SHOX deficiency, Noonan, Silver-Russell, Turner, CAH, precocious puberty (cohort 1) and mucopolysaccharidoses and other storage disorders (cohort 2); two to five expert Greulich-Pyle readings each.
- **Method:** the open-source Deeplasia system run unchanged; each human rater and the model scored against the remaining experts.
- **Result:** mean absolute error 6.0 months (cohort 1) and 7.1 months (storage disorders), 1-year accuracy 90% and 81%; the model beat every individual human rater.
- **Why it matters:** an open pediatric model validated on the hardest cases, by an outside group, on external data: the template for how a children's hospital should test any tool.

Pediatric PET at one fifth of the counts with patch-based deep learning



Han et al., EJNMMI Phys 2026, figure 5: pediatric whole-body FDG

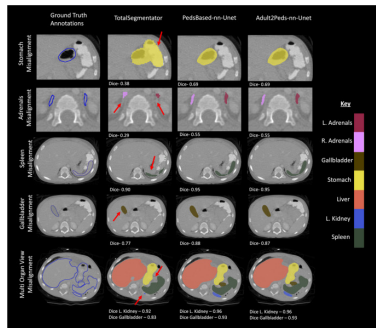
PET, standard 90 s vs 20 s per bed, with and without patch-based

deep-learning denoising

source: cdn.ncbi.nlm.nih.gov

- **Problem:** PET dose and scan time in children are set by noise; conventional denoising networks need large training sets that pediatric PET does not have.
- **Data:** one high-quality pediatric exam for training; a NEMA phantom and 88 clinical whole-body FDG exams (under 12 years) with list-mode data truncated from 90 s to 20 s per bed.
- **Method:** a patch-based network that learns local structure from a single exam and denoises the reduced-count images; blinded observer study and SUV analysis.
- **Result:** 20-second images rated equal or better than the standard 90-second images in more than 85% of cases, with resolution, contrast and SUV preserved.
- **Why it matters:** either a 4.5x shorter scan or about 22% of today's injected activity, from a method trainable on a single local exam.

Children are not small adults: an adult CT segmentation model in children



Chatterjee et al., J Imaging Inform Med 2025, figure 7: pediatric CT

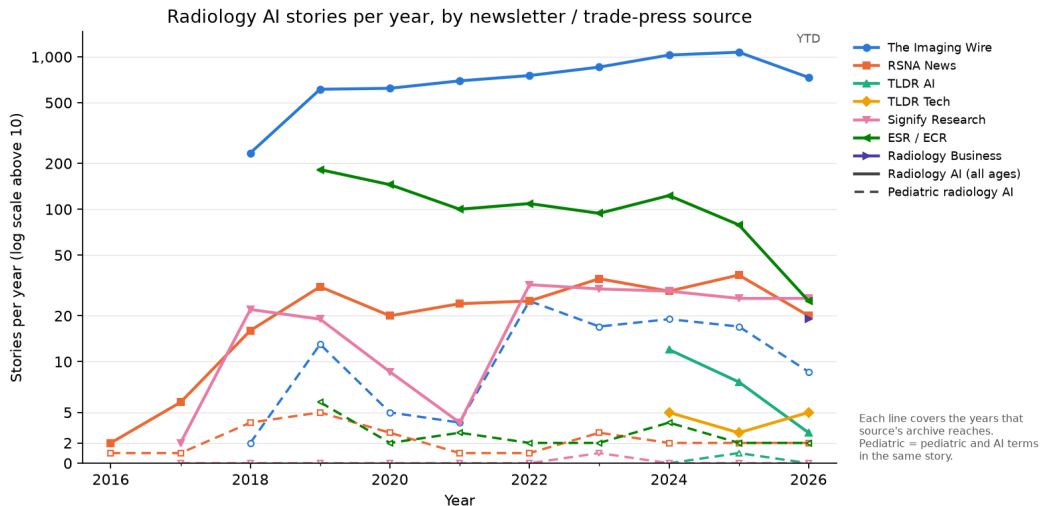
organ labels, ground truth vs adult-trained TotalSegmentator vs

pediatric-trained models

source: cdn.ncbi.nlm.nih.gov

- **Problem:** the most-used open CT organ segmentation model (TotalSegmentator) was trained on adults; nobody had measured how it behaves in children.
- **Data:** external adult ($n = 300$) and pediatric ($n = 359$) abdominal CT sets with expert organ labels.
- **Method:** TotalSegmentator scored on both sets; then a pediatric-only nnU-Net and an adult model fine-tuned on pediatric cases.
- **Result:** mean Dice fell from 0.81 in adults to 0.73 in children, worst for the adrenals (0.69 to 0.41 and 0.35), duodenum and pancreas; pediatric training or fine-tuning recovered the loss.
- **Why it matters:** the clearest quantitative case for local pediatric validation and fine-tuning of adult-derived tools before clinical use.

Newsletters — radiology AI stories per year, by source



Pediatric radiology AI in the trade press, 2023 (23 stories)

Date	Newsletter	Story	Paper
2023-01-08	The Imaging Wire	Gestational Age AI	
2023-03-01	The Imaging Wire	Fetal Ultrasound AI Generalizability	Sendra-Balcells et al., 2023 (Scientific Reports)
2023-03-08	ESR / ECR	ECR 2023 – A true celebration of the cycle of life.	
2023-03-17	The Imaging Wire	When Sao Paolo's Diagnosticos da America (DASA, the world's fourth largest diagn	
2023-04-07	The Imaging Wire	Ultrasound Spots Breech Pregnancies	Knights et al., 2023 (PLOS Medicine)
2023-04-07	The Imaging Wire	When Sao Paolo's Diagnosticos da America SA (DASA, the world's fourth largest di	
2023-04-12	Signify Research	Where do opportunities exist?	
2023-04-12	RSNA News	RSNA Webinar to Examine Impact of Large Language Models	
2023-05-07	The Imaging Wire	New C-Arm Launched	
2023-06-07	The Imaging Wire	BrightHeart's Fetal Heart AI	
2023-06-12	RSNA News	Understanding the Imaging Spectrum of non-Hodgkin Lymphoma in Children, Adolescents and Young Adults	
2023-06-19	The Imaging Wire	Better Together at SIIM	
2023-06-25	The Imaging Wire	AI Detects Child Abuse on CT	
2023-07-07	RSNA News	Survivors of Childhood Cancer: Risk of Future Disease, Screening Needs and the Role of Radiology	
2023-08-02	The Imaging Wire	MRI Radiomics Hones in on Crohn's	Liu et al., 2024 (American Journal of Roentgenology)
2023-08-02	The Imaging Wire	Sonio Gets FDA Nod for Fetal AI	
2023-09-20	The Imaging Wire	GE Lands \$44M AI Ultrasound Grant	
2023-10-08	The Imaging Wire	ChatGPT Reports for Patients	Jeblick et al., 2023 (European Radiology)
2023-11-08	The Imaging Wire	Philips to Expand AI-Powered POCUS	
2023-11-08	The Imaging Wire	Stories of Success in Cloud and AI	
2023-11-12	The Imaging Wire	Uneven Success Against Breast Cancer Pediatric Radiation	
2023-11-26	The Imaging Wire	AI of MRI Diagnoses Autism	
2023-12-11	ESR / ECR	Revolutionising paediatric radiology: the future impact of AI	

Pediatric radiology AI in the trade press, 2024 (25 stories)

Date	Newsletter	Story	Paper
2024-01-10	The Imaging Wire	AI Models Go Head-to-Head in Project AIR Study	van Leeuwen et al., 2024 (Radiology)
2024-01-15	ESR / ECR	Hidden health risks of endocrine disruptors?	
2024-01-17	The Imaging Wire	Echo AI's RHD Impact	Brown et al., 2024 (Journal of the American Heart Association)
2024-01-20	The Imaging Wire	Us2.ai	
2024-02-07	The Imaging Wire	Meet Gleamer at ECR 2024	
2024-03-20	The Imaging Wire	AI Helps Residents Detect Fractures	
2024-04-22	The Imaging Wire	GE Launches US Scanners	
2024-05-19	The Imaging Wire	Early Preeclampsia Risk Screening	Guerby et al., 2024 (Hypertension)
2024-06-29	The Imaging Wire	FDA Clears New Clarius Prenatal App	
2024-07-01	ESR / ECR	AI applications in musculoskeletal imaging	
2024-07-10	RSNA News	Experts Outline Considerations to Deploy AI in Radiology	
2024-08-14	The Imaging Wire	Cardiac Ultrasound Scanner Cleared	
2024-08-25	The Imaging Wire	When Follow-Up Falls Short Pediatric CT Use	
2024-08-25	The Imaging Wire	MAUI Emerges from Stealth	
2024-09-04	The Imaging Wire	Samsung Finalizes Sonio Acquisition	
2024-09-08	The Imaging Wire	AZmed AI Extended to Kids	
2024-10-03	ESR / ECR	On Artificial Intelligence: An interview with Susan Shelmerdine	
2024-10-06	The Imaging Wire	Fracture AI Improves Over Time	
2024-10-10	The Imaging Wire	Low-Dose CT in Kids Hinton Nabs Nobel, Cancels MRI	
2024-10-10	The Imaging Wire	Low-Dose CT Confounds CAD in Kids	Hardie et al., 2025 (American Journal of Roentgenology)
2024-10-17	ESR / ECR	On Artificial Intelligence: An interview with Brendan Kelly	
2024-11-19	RSNA News	Incorrect AI Advice Influences Diagnostic Decisions	
2024-11-25	The Imaging Wire	Fetal Ultrasound AI Cleared	
2024-12-02	The Imaging Wire	Milvue Joins deepc AI Platform	
2024-12-15	The Imaging Wire	Researchers Study AI for Pediatric Fractures	

Pediatric radiology AI in the trade press, 2025 (22 stories)

Date	Newsletter	Story	Paper
2025-01-08	The Imaging Wire	AI for Pediatric Fracture Detection	
2025-01-08	The Imaging Wire	Children's National taps Microsoft to prototype AI tools .	
2025-01-29	The Imaging Wire	Samsung Launches OB/GYN Ultrasound	
2025-02-02	The Imaging Wire	Ultrasound AI Detects Fetal Defects	
2025-02-06	ESR / ECR	Philips at #ECR2025: New AI-enabled systems, intelligent software, and imaging cloud services streamline radiology workflows, advance clinical insights	
2025-02-14	ESR / ECR	Learn more about FUJIFILM's latest innovations at ECR!	Iwan et al., 2020 (European Radiology)
2025-02-27	The Imaging Wire	AI Detects Pediatric Epilepsy	
2025-03-04	TLDR AI	AI to diagnose invisible brain abnormalities in children with epilepsy (4 minute read)	
2025-04-02	The Imaging Wire	AZmed Gets New Chest AI Clearances	
2025-04-13	The Imaging Wire	GE's Cincinnati Pediatric Partnership	
2025-05-14	The Imaging Wire	Philips Partners with NVIDIA on AI for MRI	
2025-05-18	The Imaging Wire	New GE Clearances	
2025-05-20	RSNA News	Radiologists Share Tips to Prevent AI Bias	
2025-06-01	The Imaging Wire	BrightHeart's 3rd FDA Clearance	
2025-07-03	The Imaging Wire	DeepEcho Gets FDA Nod for Fetal AI	
2025-07-10	RSNA News	How Is AI Being Used in Daily Neuroradiology Practice?	
2025-08-03	The Imaging Wire	Radiologist Pay Jumps Nearly 8% in New Survey	
2025-08-13	The Imaging Wire	AI-Assisted Fracture Detection	
2025-08-27	The Imaging Wire	Sonio Launches New Maternal-Fetal AI Options	
2025-10-01	The Imaging Wire	Carebot Expands AI Portfolio	
2025-10-08	The Imaging Wire	Reducing CT Radiation Dose System-Wide	
2025-11-09	The Imaging Wire	AI Detects Pediatric Elbow Fractures	

Pediatric radiology AI in the trade press, 2026 (year to date) (13 stories)

Date	Newsletter	Story	Paper
2026-01-07	The Imaging Wire	BrightHeart Raises EUR 11M	
2026-01-28	The Imaging Wire	AHA Publishes New Stroke Guidelines	Prabhakaran et al., 2026 (Stroke)
2026-01-28	ESR / ECR	Guerbet: Innovation and French Cooperation in the Spotlight at ECR 2026	
2026-02-01	The Imaging Wire	DL-Based CT Reduces Radiation Dose	
2026-02-08	The Imaging Wire	GE Bolsters Ultrasound with Fetal AI	
2026-02-26	ESR / ECR	MRI Unveiling at the	
2026-03-29	The Imaging Wire	Ultrasound AI Gets Breakthrough Nod	
2026-05-10	The Imaging Wire	Pediatric MRI Safety, RTs and Remote Radiology, and PSMA-PET	
2026-06-22	RSNA News	Researchers Use AI to Establish Body Charts for CT Imaging Across the Adult Lifespan	
2026-06-28	The Imaging Wire	BrightHeart's Bright Future	Zelop et al., 2025 (Obstetrics & Gynecology)
2026-07-12	The Imaging Wire	AI's Rise in Pediatric Imaging	Shah et al., 2026 (European Radiology)
2026-08-16	The Imaging Wire	CT vs. MRI for Pediatric TBI, Helium Vulnerability, and Ditching the Disk	
2026-08-25	RSNA News	AI Challenges Fuel Innovation Beyond the Leaderboard	

Commercial software with a pediatric angle

Vendor	Product	Task	Pediatric status	FDA list
Gleamer	BoneView	fracture detection on radiographs	pediatric indication (age 2+) cleared 2023	4 (2022–2025)
AZmed	Rayvolve	fracture (and chest) findings on radiographs	pediatric fracture indication cleared 2024	4 (2022–2025)
Visiana	BoneXpert	automated bone age (GP / TW)	pediatric by design; CE-marked since 2009 (not on the FDA AI list)	not listed
16 Bit	Rho	automated bone age	pediatric by design	1 (2024)
Ever Fortune.AI	EFAI Bonesuite Bone Age Pro	automated bone age	pediatric by design	11 (2022–2026)
BrightHeart	Fetal EchoScan	fetal echocardiography view / anomaly assist	prenatal (fetal) by design	5 (2024–2025)
GE HealthCare / Canon / Siemens / Philips	TrueFidelity, AiCE, Deep Resolve, Precise Image	deep-learning CT / MR reconstruction (lower dose, faster scans)	adult-cleared scanner software, widely used for pediatric dose reduction	307 (2014–2026)
Subtle Medical / AIRS Medical	SubtleMR, SwiftMR	accelerated MRI via deep-learning denoising	adult-cleared; pediatric scan-time studies	13 (2018–2026)
Qure.ai	qXR / qER	chest radiograph and head CT triage / quantification	adult indications; pediatric TB-screening evidence only	9 (2020–2026)
Aidoc / Viz.ai	BriefCase, Viz LVO / ICH	worklist triage (hemorrhage, LVO, PE, pneumothorax)	adult-only indications; the largest deployed category	43 (2018–2026)

“FDA list” = number of radiology AI-enabled devices the company has on the FDA list (years). Pediatric status is the publicly stated indication; confirm the age range in the 510(k) summary before purchase.

Worth knowing as a radiologist — and why

- **TotalSegmentator** (open tool): free, one-command segmentation of 100+ structures on any CT (and MR); the basis for opportunistic measurements (bone density, muscle, organ volumes)
- **nnU-Net** (open tool): if a vendor claims a segmentation result, this is the baseline they had to beat
- **MONAI** (open framework): what your data scientists will build with; knowing the name lets you read a methods section
- **RSNA Pediatric Bone Age Challenge (2017)** (paper / dataset): the one pediatric task with a public benchmark, many models, and cleared products (BoneXpert, Rho)
- **"Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs"** (Zech et al., PLoS Med 2018) (paper): the cleanest demonstration that a model can learn the hospital instead of the disease; the reason to demand local validation
- **CLAIM checklist (Radiology: AI 2020)** (guideline): how to tell a well-reported AI study from a weak one
- **"Improving Image Quality and Reducing Radiation Dose for Pediatric CT by Using Deep Learning Reconstruction"** (Brady et al., Radiology 2021) (paper): the most-cited non-bone-age pediatric paper: deep-learning reconstruction lowers pediatric CT dose with better image quality; the most immediately usable pediatric benefit
- **FDA AI-enabled device list** (registry): the authoritative list of what is cleared; check the indication's age range before buying
- **Gleamer BoneView / AZmed Rayvolve** (products): the first fracture-detection tools with pediatric indications; realistic first pediatric AI purchases
- **MedSAM / segment-anything models** (foundation models): where annotation and segmentation tooling is heading: click, don't draw

What the field does well

- **Finding what is there.** Detection / diagnosis is the bulk of the literature (44% of radiology-AI papers, 46% of pediatric); in adults it has become deployed worklist triage, the largest block of the 1,164 FDA-listed radiology AI devices.
- **Outlining and measuring.** Segmentation (26% of papers) has open, strong defaults (nnU-Net, TotalSegmentator); in children the mature measurement task is bone age (RSNA challenge: 4.2-month error, several cleared products), and measurement / quantification is 22% of pediatric papers.
- **Better images from less dose.** Reconstruction / imputation is 22% of radiology AI and 12% of pediatric work; deep-learning reconstruction is on today's scanners and cut pediatric CT dose by about half at equal or better image quality.
- **Shared benchmarks where they exist.** Fetal and neonatal brain MRI (FeTA, dHCP) and pediatric brain tumors (BraTS-PEDs) now have challenge datasets, and multi-site pediatric neuro-oncology models match radiologists retrospectively.

- **Foundation and vision-language models:** 7.2% of radiology-AI papers (4.0% pediatric); segment-anything style tools (MedSAM) and generalist radiology models are replacing task-specific training.
- **Report generation and LLMs:** 3.8% of papers; drafting and extracting from reports, with fluency ahead of reliability.
- **Agents:** 1.0% of radiology-AI papers (42 pediatric papers in total); LLMs that call imaging tools and take multi-step actions.
- **Pediatric-specific:** bone age robust to skeletal dysplasias (Deeplasia); fetal ultrasound AI is the most active commercial pediatric area in the trade press (BrightHeart, Sonio, DeepEcho clearances, 2023–2026); adult fracture tools extended to children (Gleamer 2023, AZmed 2024) and now tested in a real pediatric emergency department.
- **Generalization studies:** adult-trained CT organ segmentation degrades in small children, and the first pediatric prospective-style reader studies show smaller gains than stand-alone accuracy implies.

Open problems — where a children's hospital could contribute

- **Which diseases?** Since 2023 the pediatric corpus (6,812 papers) leans to autism / ADHD / neurodevelopment (710); fetal anomalies / prenatal (541); brain tumors (337); congenital heart disease / cardiac (302). Barely studied: testicular / ovarian torsion (8); inflammatory bowel disease (23); lines and tubes / NICU support devices (54); neonatal jaundice / biliary atresia / liver (56); craniosynostosis / craniofacial (72); chronic lung disease of prematurity / neonatal lung (74). Which of these deserve a model first is an open question for this group.
- **Dataset curation.** What would a multi-center pediatric dataset look like: age-stratified, consented, protocol-diverse, with rare phenotypes? Which of our own archives (CT, ultrasound, NICU radiographs) could become the pediatric benchmark that does not yet exist?
- **Validation.** Local, age-stratified testing of adult-cleared tools; prospective reader studies rather than retrospective accuracy; monitoring for drift as children grow and protocols change.
- **Beyond interpretation.** Workflow / non-interpretive work is 13% of papers, but communication with families, education, protocoling and policy are almost absent; these are cheaper to build and evaluate than diagnostic models.
- **Regulation and liability.** Few of the 1,164 radiology clearances carry a pediatric indication; which adult clearances are acceptable to use off-label in children, and under what local validation?

Implications for a children's hospital

- 1 **Buy maturity, build for the gaps:** adopt cleared adult-derived tools that transfer (reconstruction, bone age, fracture with a pediatric indication); treat the under-studied pediatric problems as local validation and research.
- 2 **Demand local pediatric validation** before clinical use, by age group, and check the age range in the clearance.
- 3 **Prioritize dose and throughput:** deep-learning reconstruction gives the clearest pediatric benefit today.
- 4 **Invest in data:** a curated, shareable pediatric dataset is the scarcest resource in the field and the contribution a children's hospital is uniquely placed to make.